Experiment 10: Intelligent Decision Support System

Aim

To develop an intelligent decision-support system using rule-based reasoning, learning from examples, and ethical AI evaluation.

Exercises:

1. **Expert System:** Develop a rule-based system with a knowledge base and inference engine to generate recommendations for a selected application.
2. **Imitation Learning:** Use example decisions provided by a human to train a model to imitate those decisions and evaluate its predictions.
3. **Explanation Facility:** Display the rules or input factors responsible for the system's recommendations.
4. **Ethical AI:** Test the system using different groups or input conditions to identify possible bias. Evaluate fairness and transparency in its recommendations.
5. **Testing and Comparison:** Modify rules or training examples and compare the system's recommendations before and after the changes.